BENUDHAR BEHERA

Lead Backend Engineer | Java, Distributed Systems, Cloud & AI Platforms

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PROFESSIONAL SUMMARY

Lead Backend Engineer with 9 years of experience architecting and delivering scalable, resilient distributed systems and cloud-native microservices. Deep expertise in Java, Spring Boot, Kafka, Python, FastAPI, and AWS, with hands-on experience building GenAI platforms using RAG, LLMs, Vector Databases, and Model Context Protocol (MCP). Proven track record designing event-driven architectures, high-throughput APIs, and AI-enabled backend platforms with a strong focus on scalability, reliability, and clean architecture. Experienced in Agile/Scrum delivery with cross-functional collaboration across product, AI, and data engineering teams.

TECHNICAL SKILLS

Programming Languages	Java (Primary), Python
Backend & Frameworks	Spring Boot, Spring WebFlux, Spring Cloud, FastAPI, REST APIs, Microservices
AI & GenAI	RAG, LLMs, OpenAI API, Spring AI, Model Context Protocol (MCP), AI Agents, Vector Databases, LangChain
Distributed Systems	Kafka, Event-Driven Architecture, gRPC, SQS, SNS
Database & Storage	PostgreSQL, MongoDB, DynamoDB, Redis, Elasticsearch
Cloud & DevOps	AWS (EC2, S3, RDS, EKS, Lambda, SQS, SNS, CloudWatch), Terraform, Docker, Kubernetes, Jenkins, CI/CD
Architecture	Distributed Systems, Hexagonal Architecture, Microservices, Cloud-Native, System Design, Resilience Patterns
Security & Monitoring	OAuth2, Keycloak, Prometheus, Grafana, ELK Stack, OpenTelemetry, Splunk, API Rate Limiting
Design Principles	HLD, LLD, SOLID, Design Patterns, Scalability, Rate Limiting, Clean Architecture
Process & Delivery	Agile, Scrum, Sprint Planning, Stakeholder Management, Cross-functional Collaboration

KEY ACHIEVEMENTS

- **GenAI Platform at Scale – EY:** Architected a production-grade GenAI conversational platform at EY sustaining 600+ requests/minute at low latency, supporting real-time, secure, and highly available document retrieval and AI tool integration.
- **Performance Turnaround – Wipro:** Improved API response times by 40%+ through strategic caching, JVM tuning, and database query optimization on a large-scale enterprise platform, significantly enhancing system throughput.
- **Reliability at Scale – ITC InfoTech:** Engineered idempotent and failure-resilient payment workflows processing 10,000+ daily transactions with zero duplicate transaction failures across a live e-commerce order pipeline.
- **Enterprise Application – Silicon Tech Lab:** Developed and delivered ERP automation modules reducing manual business-process effort by ~30%, deployed across 5+ government departments supporting ~5,000 users from day one.

PROFESSIONAL EXPERIENCE

Lead Backend Engineer (Senior Technology Consultant) | Ernst & Young (EY)

Bangalore, India • Mar 2024 – Present

- Architected and delivered a production-grade GenAI conversational platform integrating LLMs with large-scale research datasets, supporting real-time, secure, and highly available document retrieval sustaining 600+ requests/minute at low latency.
- Engineered cloud-native microservices using Java 21, Spring Boot, and Kafka following Hexagonal Architecture, enabling scalable, maintainable, and testable enterprise applications.

- Built RAG-based knowledge retrieval systems using vector embeddings and semantic search, improving contextual response accuracy and retrieval efficiency across large-scale datasets.
- Designed MCP-based AI tool integrations enabling LLMs to dynamically discover and invoke backend capabilities, connecting conversational AI workflows with enterprise services and data.
- Reduced API latency by ~40% through caching, query optimization, JVM tuning, and asynchronous processing, improving system throughput and responsiveness under high load.
- Partnered with product managers, AI engineers, and data teams using Agile/Scrum practices to deliver secure, production-ready AI solutions on time and at scale.

Senior Backend Engineer (Senior Project Engineer) | **Wipro Limited**

Bangalore, India • Jul 2021 – Feb 2024

- Built cloud-native microservices using Java, Spring Boot, and Kafka for a large-scale enterprise platform, enabling scalable backend services and real-time event processing.
- Improved API response time by 40%+ through query optimization, caching strategies, asynchronous processing, and JVM tuning, significantly reducing latency under high-volume workloads.
- Optimized database indexing strategies and SQL execution plans to improve throughput for high-volume transactional workloads on PostgreSQL.
- Developed comprehensive unit and integration test suites using JUnit 5 and Mockito, improving application stability and reducing production defects.
- Resolved production incidents using Splunk, Grafana, and AWS CloudWatch, reducing mean time to resolution through proactive monitoring and log analysis.

Backend Engineer (Associate IT Consultant) | **ITC InfoTech**

Bangalore, India • Sep 2020 – Jul 2021

- Developed scalable cloud-native backend services using Java, Spring Boot, and RESTful APIs for an enterprise e-commerce platform, supporting order lifecycle management and real-time payment processing.
- Implemented event-driven architecture using Kafka to support asynchronous order processing, payment notifications, and inventory synchronization across distributed services.
- Designed idempotent APIs and resilient payment workflows using retry and failure-recovery mechanisms, ensuring zero duplicate transaction failures across 10,000+ daily transactions.
- Built reusable business components following SOLID principles and clean architecture, improving code maintainability and reducing onboarding time for new engineers.
- Increased test coverage and production stability through comprehensive automated testing using JUnit 5 and Mockito.

Software Engineer | **Silicon Tech Lab Pvt Ltd.**

Bhubaneswar, India • Nov 2017 – Sep 2020

- Developed backend modules using Java, Spring MVC, Hibernate ORM, and PostgreSQL for enterprise ERP applications supporting business process automation across government departments.
- Delivered production-ready ERP features supporting ~5,000 users across 5+ government departments, reducing manual business-process effort by ~30%.
- Resolved production issues through root cause analysis, bug fixing, and performance tuning, improving application stability and customer satisfaction.
- Collaborated with senior engineers, QA teams, and business analysts to deliver production-ready features following Agile development practices.

CERTIFICATIONS

- EY IT Architecture – Solution Architecture (Bronze), 2025
- EY Artificial Intelligence – Applied AI (Bronze), 2024
- Introduction to Generative AI – Google Cloud

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science & Engineering

Silicon Institute of Technology, Bhubaneswar | 2013 – 2017 | CGPA: 7.2 / 10